

2025-26

Time : 3 hours

Full Marks : 75

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

(Compulsory)

1. Answer **all** questions of the following : $1 \times 5 = 5$
 - (a) What is meant by stopping potential in the photoelectric effect ?
 - (b) What does the Davission and Germer experiment demonstrate ?
 - (c) Write down the expression for radius of a nucleus.

- (d) How does the volume term in the semi-empirical mass formula contribute to binding energy?
- (e) Under what condition does population inversion take place in a laser medium.
2. A particle is described by the one-dimensional wave function. 5
- $$\psi(x) = A e^{-\alpha x^2}, \quad -\infty < x < \infty,$$
- where A is the normalization constant and α is a positive constant. Write down the normalization condition for the wave function. Hence determine the value of the normalization constant A in terms of α .
3. Differentiate between spontaneous emission and stimulated emission, highlighting the role of an incident photon in each process. 5

Group – B

Answer any **four** questions of the following :

$$15 \times 4 = 60$$

4. A particle of mass m is confined in a one-dimensional infinite square potential well of width L , such that the potential $V(x) = 0$ for $0 < x < L$

and $V(x) = \infty$ elsewhere. Write the time-independent Schrodinger equation for the particle inside the well and solve it obtain the allowed energy eigenvalues. Write the normalized eigenfunctions corresponding to these energy levels.

5. Explain the Heisenberg gamma-ray microscope throught experiment in detail and discuss how it leads to the Heisenberg uncertainty principle, highlighting the limitations imposed on the simultaneous measurement of an electron's position and momentum.
6. Describe Fermi theory of Beta Decay and explain why neutrino are needed in the hypothesis.
7. Explain Gamow's theory of alpha decay. Describe the quantum mechanical tunneling mechanism involved in alpha emission from a nucleus and illustrate the process with the help of a potential energy barrier diagram (graph), explaining how the alpha particle penetrates the Coulomb barrier despite having insufficient classical energy.

8. Derive the time-dependent Schrödinger equation starting from the concept of matter waves and hence obtain the time-independent Schrödinger equation by the method of separation of variables.
9. Write short notes on any **two** of the following :
- (a) Define half-life and derive the expression for the half-life of a radioactive substance.
 - (b) Derive and explain the rate equations for population densities in a two-level laser system and discuss why population inversion cannot be achieved under steady-state conditions.
 - (c) Physical interpretation of wave function.
 - (d) de-Broglie's hypothesis on wave-particle duality and explain its physical significance.

